Training a Model to Determine Playoff Results in the NHL

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Abstract

The project leverages data from the National Hockey League (NHL) to analyze and determine patterns and trends in both regular season and postseason performance. The central aim is to identify which regular season metrics are most predictive of playoff success. Given the observed unpredictability of the NHL playoffs, where the most dominant regular season team often fails to to win the Stanley Cup, we set out to construct a machine learning model capable of improving postseason predictions.
After experimenting with neural network and random forest classifier, we ultimately decided to chose logistic regression for its clarity, interpretability, and relative effectiveness. By classifying all 32 teams into four postseason outcomes, non-playoff, playoff participant, conference finalist, and Stanley Cup champion, we sought to to balance model accuracy with meaningful insight. The project was conducted in Python using Google Colab, where we collaborated on data preprocessing, model development, and results visualization. While no model can eliminate the chaos inherent in playoff hockey, our approach demonstrates that meaningful, accurate predictions can still emerge through careful model selection, feature engineering, and class balancing strategies.

Division of Work

To maintain efficiency throughout the research implementation process, the project was divided according to our strengths. Madison took the lead in conceptual design, conducting preliminary research, identifying relevant statistical features, and shaping the theoretical direction of the study. She also documented the process throughout, recording insights, challenges, evolving methodologies.
Anthony focused on the technical side, including sourcing and preprocessing the data from NHL.com, ensuring it was clean, structured, and aligned with our goals. He implemented the machine learning models,

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managed testing workflows, and fine-tuned model performance. Both of us collaborated in ensuring code accessibility, aiming to create a replicable and expandable framework for future researchers interested in sports analytics.

Introduction

While machine learning has made significant inroads into sports analytics in recent years, the NHL presents a particularly intriguing challenge. The leagues high degree of parity, combined with the variability of single-elimination series, often leads to unexpected playoff outcomes. A study we reviewed indicated that the top regular season team only wins the stanley cup about 18-20% of the time [4], a far cry from predictive certainty.

Previous models have attempted various techniques, with support vector machines (SVM) showing promise in some cases. However, many of these models struggled to rise above 65% accuracy when tasked with postseason prediction [2]. After reviewing existing literature and considering the complexity of our classification task, we opted to use logistic regression to create a four-tiered classification model for team outcomes: 0 = no playoffs, 1 = playoff appearance, 2 = conference finalist, 3 = Stanley Cup winner.

This choice allowed us to maintain clarity while emphasizing interpretability ove black-box complexity. Our goal was to provide more than just raw predictions, we wanted a framework that offered insight into the dynamics of playoff qualification and deep postseason advancement.

Background

The use of machine learning in sports has grown dramatically, from player performance metrics to full-game outcome predictions. In the NHL, specifically, parity and randomness in short playoff series pose unique challenges. Over five seasons of data, we observed that traditional statistical indicators, such as win percentage or goal differential, often failed to constantly predict postseason success. The observation underscores the need for models that can capture nonlinear interactions and underrepresented classes without overfitting. Initial testing with neural network showed potential but lacked stability and

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interpretability. Our random forest model performed poorly, classifying every team as a playoff contender. These shortcomings revealed a core issue: class imbalance. Only a few teams make deep postseason runs each year, typically just four conference finalists and one Stanley Cup winner. Many algorithms failed to account for these skewed distributions. Logistic regression, with manual and programmatic class weighting, gave us more flexibility to tune our model to reflect the actual structure of NHL postseason outcomes.
Additionally, logistic regression offers clearer insight into which variables matter most. While the best model is not always the most interpretable, in sports analytics, where fans, teams, and analysts care about understanding why a team might succeed, transparency is essential.

Methodology

Our model classified teams based on regular season statistics into one of four postseason categories. After discarding neural network and random forest approaches due to their performance issues, we settled on multinomial logistic regression. This model was tuned to weigh each class more appropriately, compensating for the small number of Stanley Cup winners and conference finalists relative to playoff qualifiers and non-qualifiers. Key features included were points percentage (Pts%), Regulation wins, goal differential, special teams efficiency (power play and penalty kill percentages), save percentage, and Corsi and Fenwick measures (to capture puck possession). The data collected from NHL.com and adjusted slightly to reflect playoff performance. All modeling was conducted in Google Colab using Python, employing libraries such as pandas, scikit-learn, and seaborn for data processing, visualization, and model development.
Because the classes were inherently imbalance, we tested three weighting approaches, equal class weights, weighting by inverse frequency, and manually tuned weights to increase the predictive likelihood for classes 2 and 3. Our final model used the third approach. By iterating on weights and thresholds, we improved our ability to predict rare outcomes without sacrificing accuracy for common cases.

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Experimentation

We tested our model on historical seasons to evaluate predictive performance. Initial results from early models (equal weights, random forest, neural net) showed the weakness in capturing lower-frequency classes. After manual adjustment, the logistic regression model began to provide more realistic class distribution predictions. Our primary testing season was 2024. When trained on earlier data and tested on 2024 standings, the model correctly identified 11 of the 16 playoff teams, a 69% success rate, exceeding the typical 60-65% benchmark reported in the literature [1] [2].
We also ran a “near-real-time” test using standings from early march 2025. Most teams had played between 65 and 72 games. Here, the model correctly identified 15 of the 16 playoff teams, yielding a 94% success rate. Most notably, the model predicted the Winnipeg Jets as the likely Stanley Cup champion, a prediction that aligned with their third-best betting odds according to ESPN.com This alignment suggested not just accurate prediction but competitive insight.

Results

Our final model showed strong predictive power for playoff qualification and moderate performance in identifying deeper playoff outcomes. Predicting conference finalists and Stanley Cup winners remains difficult due to inherent randomness and limited data points, but our adjusted weights and probabilistic thresholding strategies improved accuracy beyond baseline expectations.

The Winnipeg Jets prediction was an example of the model capturing deeper postseason potential through statistical signal rather than seeding or reputation alone. Further analysis revealed that their strong underlying metrics—such as goal differential, regulation win percentage, and special teams efficiency—were significant predictors in our regression coefficients.

One clear takeaway from this project is the power of combining domain knowledge with machine learning. Understanding the NHL's structure and playoff volatility allowed us to build a model that wasn’t just accurate—it was meaningful. It provided actionable insights, such as identifying

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which performance indicators mattered most for long-term success.

Conclusion

This project explored the potential for using machine learning to predict NHL postseason outcomes, with an emphasis on logistic regression as a tool for classification. We demonstrated that, even in a sport known for its playoff chaos, a well-structured model using thoughtful weighting and feature selection can deliver valuable predictions.

Our final model achieved a 94% success rate in predicting playoff teams from late-season standings and aligned closely with expert betting markets in its Stanley Cup pick. While room for improvement remains—particularly in identifying deep playoff runs—the foundation we've laid is promising. Future iterations could benefit from additional features such as injury data, strength of schedule, and advanced possession metrics. Moreover, combining logistic regression with ensemble or hybrid models may offer even greater predictive accuracy.

Ultimately, our work highlights the growing potential of machine learning in sports analytics. With thoughtful application, even relatively simple models can illuminate the factors that drive postseason success and bring statistical rigor to one of hockey's most unpredictable stages.

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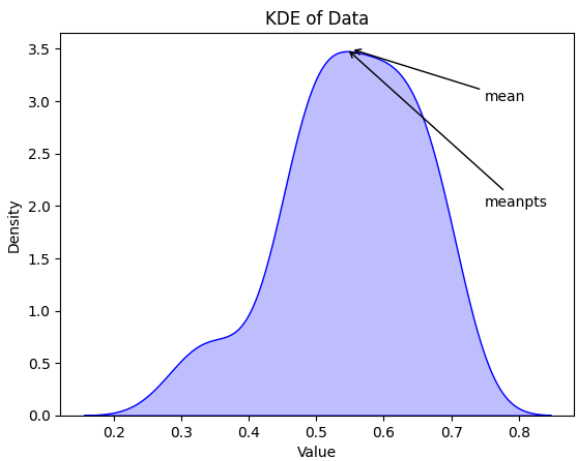
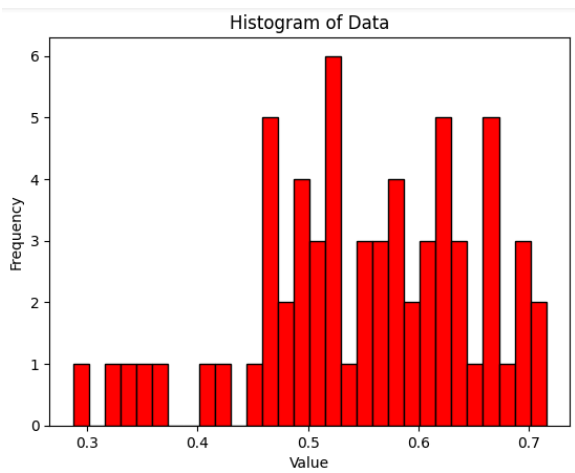
Appendix

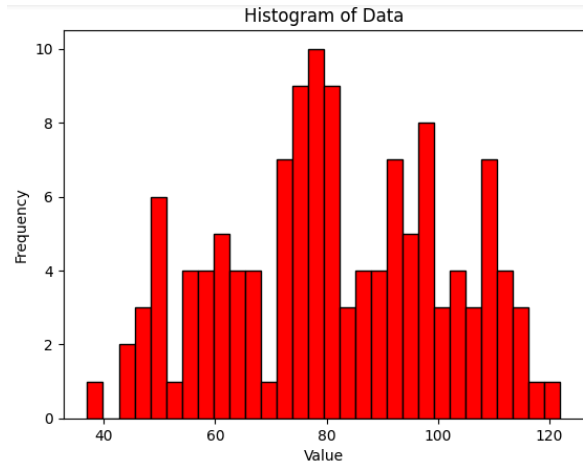
Note: most of these graphs were used to figure out the modality of the data to be able to do tests and train a model on it without any inherent biases present in the original training data



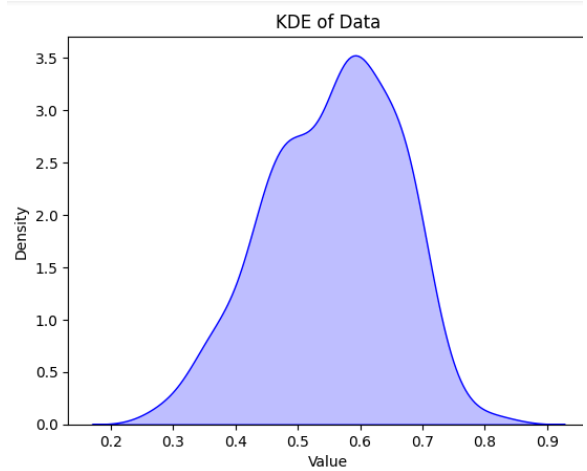
Graph 1: a KDE graph of the pts% of teams over two seasons, similar to a smoothed-out histogram



Graph 2: histogram showing the distribution of pts% over the past two seasons, seams like it could possibly be multimodal, or at the least a left skew



Graph 3: a histogram of NHL teams' points over the past 5 years, showing how there is a distinct center



Graph 4: KDE graph of the same data, showing how over time the distribution seems to be normal with a center around .550 or around 90 points rather than the expected 82 points (41-41, or exactly split) which could be due to overtime losses counting as one point (for example 41-33-8 is 90 points, but still technically 41 losses)

Github Link:

https://github.com/arholleran/csci334project

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Related Papers:

[1]Weissbock, Joshua. *Forecasting success in the National Hockey League using in-game statistics and textual data*. Diss. Université d'Ottawa/University of Ottawa, 2014.

[2]Pischedda, Gianni. “Predicting NHL Match Outcomes with ML Models.” International Journal of Computer Applications 101.9 (2014): 15–22. Web.

[3]Drever, Zack. *NHLPostSeasonNNs*. GitHub, 2023, https://github.com/zdrever/NHLPostSeasonNNs.

[4]Favero, Gian. *NHL-Postseason-Predictor*. GitHub, 2023, https://github.com/faverog/NHL-Postseason-Predictor.

[5]Orsi, Micaela, et al. "Expert System for Ice Hockey Game Prediction: Data Mining with Human Judgment." *International Journal of Modern Physics C*, vol. 27, no. 10, 2016, p. 1640002, https://doi.org/10.1142/S0219622016400022.

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